

$$1. S \rightarrow 0AO \mid 1B1 \mid BB$$

$$A \rightarrow C$$

$$B \rightarrow S \mid A$$

$$C \rightarrow S \mid \epsilon$$

a) Elimine las producciones ϵ .

$\{S, A, B, C\}$ son anulables porque $C \rightarrow \epsilon$, $A \rightarrow \epsilon$, $B \rightarrow A$ y $S \xrightarrow{*} A$.

$$\begin{array}{l|l} S \rightarrow 0AO \Rightarrow S \rightarrow 0AO \mid 00 & B \rightarrow S \mid A \\ S \rightarrow 1B1 \Rightarrow S \rightarrow 1B1 \mid 11 & A \rightarrow C \\ S \rightarrow BB \Rightarrow S \rightarrow BB \mid B \mid \epsilon & C \rightarrow S \end{array}$$

Generamos otro símbolo inicial por $S \rightarrow \epsilon$.

$$S_{\phi} \rightarrow S \mid \epsilon$$

y nos quedamos $S \rightarrow \epsilon$

b) Producciones unitarias.

$$A \Rightarrow C \text{ y } C \Rightarrow S. \quad A \Rightarrow S \parallel S \rightarrow 0AO \mid 00 \mid 1B1 \mid 11 \mid BB \mid \epsilon$$

$$A \Rightarrow S \text{ y } B \Rightarrow S \mid A \Rightarrow B \Rightarrow S \mid S \mid B = B \Rightarrow S.$$

$$C \mid A \mid B \rightarrow S = C \mid A \mid B \Rightarrow 0AO \mid 00 \mid 1B1 \mid 11 \mid BB$$

$$\text{Y } S_{\phi} \rightarrow \epsilon \mid 0AO \mid 00 \mid 1B1 \mid 11 \mid BB$$

c)

a) Por 00 y 11 todas son generadoras no se eliminan.

b) C es inalcanzable. X

$$S_{\phi} \rightarrow \epsilon \mid 0AO \mid 00 \mid 1B1 \mid 11 \mid BB \text{ y } S/A/B \rightarrow 0AO \mid 00 \mid 1B1 \mid 11 \mid BB$$

$$2. S \rightarrow aAa | bBb | \epsilon$$

$$A \rightarrow C | a$$

$$B \rightarrow C | b$$

$$C \rightarrow CDE | \epsilon$$

$$D \rightarrow A | B | ab$$

$\{S, C, A, B, D\}$ son variables

a) $S \rightarrow S | \epsilon$
 $C \rightarrow CDE | \epsilon | CE | DE, A \rightarrow C | a, B \rightarrow C | b, D \rightarrow A | B | ab, S \rightarrow aAa | bBb | aAa | bBb$

b) $C \rightarrow \epsilon$

$$A \rightarrow C$$

$S \rightarrow S | \epsilon, S \rightarrow aAa | aAa | bBb | bBb, D \rightarrow a | b | CDE | DE | CE, A \rightarrow a, B \rightarrow b, C | A | B \rightarrow CDE | DE | CE$

c)

A) ϵ no produce ϵ Todo Contiene ϵ CX

$S \rightarrow S | \epsilon, S \rightarrow aAa | aAa | bBb | bBb, A \rightarrow a, B \rightarrow b, D \rightarrow a | b | ab$

B) no alcanzables solo D. DX

$S \rightarrow S | \epsilon, S \rightarrow aAa | aAa | bBb | bBb, A \rightarrow a, B \rightarrow b$